

Geography 8.01 Introduction to International Development
Environmental Governance and Climate Action in Bangladesh
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Summary Essay

I have spent the term researching and discussing environmental governance and climate action in Bangladesh. I examined the issue primarily through the lens of SDG 13, Climate Action, specifically subtopic 13.2, “Integrate climate change measures into national policies, strategies, and planning,” and 13.3, “improve education, awareness-raising, and human and institutional capacity on climate change mitigation, adaptation, impact reduction, and early warning” (United Nations 2026). In specific posts, I also examined this vast topic through the lens of SDG 5: Gender Equality and SDG 7: Affordable and Clean Energy in environmental governance, specifically target 7.2, “by 2030, increase the share of renewable energy substantially in the global energy mix” (United Nations 2026). Overall, I have been surprised by the quality and quantity of development interventions and projects that focus on environmental governance and climate action.

After examining over 15 forms of interventions, all focusing on different subsets of sustainable development, there is no obvious panacea. However, some trends make some interventions stronger, more successful, and productive than others. Solutions that aimed to fix an existing large development issue sustainably proved to be the most effective. For example, the SHS program, a nationwide off-grid electrification program aiming to enable rural households to purchase solar photovoltaic systems for their home as a replacement for kerosene lighting, addressed an existing technological and health problem with a sustainable, cost-effective solution (Hellqvist and Heubbaum 2023). Furthermore, programs that mix technological improvements with education also seemed more effective. An example of this was the introduction of water-efficient technologies as a development project to improve sustainable agriculture in Bangladesh; the programs amplify the effects of both the new technologies and social programs,

teaching the farms to utilize the technology (SAF 2025). Conversely, interventions that seemed ineffective or misleading often stemmed from grand government documents with little proof that action had and would actually take place. An example of this would be the gender goals National Adaptation Plan of Bangladesh and the Bangladesh Climate Change and Gender Action Plan. While I was able to find the IC-DRM project in Bhola, which is actively encouraging women and girls' participation in climate risk reduction and preparedness, I failed to find any larger-scale national programs that the government documents seemed to be suggesting would be created.

Bearing these considerations in mind, the answer formula for a successful development intervention focused on environmental governance and climate action in Bangladesh is to have a government-run program that is funded in partnership with an IGO or NGO that encourages action and forces accountability. Often, these projects were run or had some partnership with the Infrastructure Development Company Limited (IDCOL), which is a government-owned development and infrastructure company founded in 1997 (IDCOL “About IDCOL”). For example, the SHS program was run primarily by the IDCOL but sourced large portions of its funding from the World Bank and the Asian Development Bank (Cabraal et al. 2021). The Solar Mini-Grids Project, the other solar intervention I discussed, was also extremely effective, and funding an implementation structure was the same as SHS. The Solar Mini-Grids Project was run by the IDCOL while getting much of its funding from the World Bank, the United States Agency for International Development (USAID), and the Asian Development Bank (IDCOL “Solar Mini-Grid”).

The dismantling of USAID will most likely have a small effect on Bangladesh's environmental governance and climate action. Many of the programs that I have examined, for example, the Solar Mini-Grids Project, have received funding from USAID, but the majority of

the funding for these programs stems from the World Bank and the Asian Development Bank. However, the ability of the World Bank and Asian Development Bank to continue to give Bangladesh so much development aid may decrease as the United States and USAID step away from international aid. Therefore, Bangladesh, while at minor risk of feeling the initial effects of the USAID cuts, is at a large risk of feeling the rippling effects across the remaining development aid economy.

The main area in which I believe more work and resources need to be devoted is social programs focused on climate action that do not pertain to cyclone and other disaster risk. In my research, I found many educational programs, like Luetz and Sultana's paper, explaining how climate change education is a key to successful disaster preparation and resilience in Bangladesh, and gender equality programs, like the IC-DRM project in Bhola, focused on climate risk and disaster preparedness. While risk is a large issue that is increasing as disasters become more frequent because of climate change, it is not the only area in which education and gender equality interventions should be utilized. Programs like the Green Belt Movement in Kenya demonstrate how education, gender development, and climate action can be threaded together to create powerful bottom-up development (Green Belt Movement). I was unable to find any multidisciplinary educational development initiatives in Bangladesh that would be comparable to the Green Belt Movement. The closest I was able to find was the combination of education and technology, during the introduction of water-efficient technologies. More programs that focus on educating and promoting women's roles in decision-making processes, while simultaneously focusing on climate action, should be created in the future.

Bangladesh is currently in a position to make significant and effective progress towards all of the SDGs I have examined. While the current ripple effects of the dismantling of USAID

are unknown, on the 17th of February 2026, Tarique Rahman was sworn in as the new prime minister of Bangladesh. This was the first election since the large student-led protest and eventual regime change that took place in 2024 (Al Jazeera 2026). This new regime is in a unique position to take power and continue to go full steam ahead in sustainable development. A focus on education and women's roles in environmental action would allow Bangladesh to not only be a leading developing nation but a nation in general when it comes to environmental governance and climate action.

Citations:

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Introducing Water-Efficient Technology in Barind Tract (IWET): Project Factsheet.

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SDG Proposal

For my SDG project, I want to examine Environmental Governance and climate action in Bangladesh. Bangladesh is currently considered a least developed country (LDC), but its economic development is growing in the post-pandemic age. I will start by looking at MDG 7 and eventually how that transitioned to SDG 13, Climate Action. I believe that I will mostly focus on subtopic 13.2 (integrate climate change measures into national policies, strategies, and planning) and 13.3 (improve education, awareness-raising, and human and institutional capacity on climate change mitigation, adaptation, impact reduction, and early warning). Still, it will also be critical to examine aspects of SDG 16 when discussing the connection between personal freedoms and environmental action. Through a brief period of research, I have already identified several specific topics that I wish to write about: climate education initiatives, governmental policies promoting environmental action, sustainable agricultural practices and plans, and nature-based solutions aimed at achieving the targets outlined in SDG 13. Collected below is a short list of sources that I have already found that will help me write posts on the aforementioned topics.

Interesting sources I had found as of posting my proposal:

Peer Review:

Ahmed, B. (2019). Environmental governance and sustainable development in Bangladesh: Millennium Development Goals and Sustainable Development Goals. *Asia Pacific Journal of Public Administration*, 41(4), 237–245.
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Background and Context

Bangladesh is a least developed country located in the Bay of Bengal in Southern Asia (Economy.com 2025). The country has a population of 173.6 million, and the official language of the nation is Bengali (World Bank 2025). With a Gross Domestic Product (GDP) per capita of 2.59 thousand United States Dollars (USD) and a GDP growth rate of 4.22%, as of 2024, the country is classified as a lower-middle-income country (World Bank CITE). The economy relies heavily on the garment industry, which accounts for more than 80% of total exports, surpassing 40 billion USD in recent years (Economy.com 2025; Paul and Johora 2025). However, recent civil unrest and tariffs by the United States (US) have dramatically hurt the economy. In the summer of 2024, mass student-run protests resulted in violent clashes with the military, which led to the eventual resignation of then-Prime Minister Sheikh Hasina. The protests known as the Non-Cooperation Movement and later the July Revolution were a pro-democratic movement initially started in response to the High Court Division of the Bangladeshi Supreme Court decision to restore quotas in government jobs. The young higher educated workforce was protesting for a true democracy and higher paying higher educated jobs outside of the garment industry (The New York Times 2024). In August of 2025, Bangladesh negotiated the initially suggested 37% tariff on exports to the US down to 20% (Paul and Johora 2025). This economic and social tension will be put to the test on February 12th when the country will head to the polls for the country's first election since former Prime Minister Hasina was forced into resignation (Shankar 2026).

Another important subject when discussing the development of Bangladesh is the 1982 debt crisis. Bangladesh went through dramatic structural adjustment brought about by the World Bank and International Monetary Fund (IMF) after the debt crisis (Sobhan 1993). The structural

adjustment was characterized by import liberalization, exchange-rate adjustments, financial deregulation, privatization, and large-scale subsidy removal. The World Bank and IMF's goal of structural adjustment was to create market-led growth, privatize industries, and promote exports, but in actuality, they just further slowed the economic growth of Bangladesh. For two decades leading up to the debt crisis and structural adjustment, Bangladesh was averaging between 5% to 7% GDP annual growth rate. The World Bank and IMF structural adjustment programs (SAPs) decreased this rate to between 3% to 4% for the following two decades (Sobhan 1993). The SAPs particularly hurt the agriculture sector, which I will be highlighting in my next post, as it is an area where climate-based solutions have been implemented to improve development. After the debt crisis and the SAPs implementation, the annual growth rate of the agricultural sector of Bangladesh fell from 4% to 1.6%, which was notably below the population growth rate, illustrating the country's new need to import food (Sobhan 1993). The SAPs also resulted in the government divesting hundreds of state-owned enterprises. Some of these were sold, but many simply shut down. The privatization was based on the assumption that private ownership would result in efficiency, but in practice, this was false (Sobhan 1993). One notable positive of the SAPs was the large boost to the garment industry. The country was encouraged to invest in the sector as it possessed a comparative advantage in producing clothing (The New York Times 2024). Garment exports soared, which created new jobs and economic possibilities, but the industry still heavily relied on imported products to produce the clothing (Sobhan 1993). There was a focus on horizontal integration and a lack of vertical integration, which decreased net gains from the comparative advantage (Sobhan 1993). Overall, the SAPs and general input from the World Bank and IMF had mainly negative effects that worsened Bangladesh's debt problems, with small positives in the advancement of the garment industry.

As I mentioned in my previous post, I will examine Environmental Governance and climate action in Bangladesh, specifically examining climate education initiatives, governmental policies promoting environmental action, sustainable agricultural practices and plans, and nature-based solutions aimed at achieving the targets outlined in Sustainable Development Goal (SDG) 13. It will also be critical to examine aspects of SDG 16 when discussing the connection between personal freedoms and environmental action. Therefore, it is important to understand the current progress that Bangladesh has made towards achieving both of these SDGs. According to the 2025 Sustainable Development Report, Bangladesh has achieved SDG 13; however, progress towards improving has been stagnant in recent years. The tons of CO₂ per capita emitted from fossil fuel combustion, cement production, and greenhouse gas emissions embodied in imports have been stagnant in recent years (Sachs et al. 2025). The same report states that major challenges remain in achieving SDG 16, with decreasing progress in recent years. Notably, the corruption perception index and press freedom index have both been either steady at low levels or decreased in the last decade. Bangladesh is actively improving its development in SDG 13, and the upcoming election results will have a direct impact on SDG 16 indicators.

Citations:

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NGO Driven Market-Based Solutions vs. Technology-Based Solutions in Sustainable Agriculture in Bangladesh

In this post, I am going to compare two programs that the Sustainable Agriculture Foundation (SAF) has run in Bangladesh to promote sustainable agriculture. The first intervention, introducing water-efficient technology, involves importing more advanced technologies to promote sustainable agriculture and improve farming practices and productivity. The second intervention, promoting risk mitigation measures for climate change adaptation, is an example of a market-based solution attempting to promote sustainable agriculture.

I first want to provide an overview of sustainable agriculture in Bangladesh. Faroque et al.'s 2011 article, "Sustainable Agriculture: A Challenge in Bangladesh," examines the challenges facing conventional agriculture in Bangladesh and argues that sustainable agricultural systems are essential for long-term productivity, economic success, social equity, and environmental health. The article highlights that sustainable agriculture requires four types of sustainability to go along with the aforementioned goals: productivity sustainability, economic sustainability, social sustainability, and environmental sustainability (Faroque et al. 2011). The term sustainable agriculture is not purely about protecting the environment, but rather how improved agricultural practices will protect the climate while simultaneously being economically and socially productive (Faroque et al. 2011). In Bangladesh, the goal of sustainable agriculture is to reduce land and water degradation, improve crop yields, and promote long-term food security and environmental quality for the nation (Faroque et al. 2011).

The first intervention is introducing water-efficient technology. Funded by the Coca-Cola foundation the program ran from 2018 to 2025 and reached 40,000 farmers. The program's goal was to "enhance agro-water productivity, reduce groundwater extraction and increase farmers'

income,” (SAF 2025). To meet these goals, SAF instructed farmers to use ultra-high-density plantations of mango and malta by using drip irrigation technology and to make use of paddy cultivation using the Alternative Wetting and Drying method. In this intervention, new technologies were the key to success, representing a belief that, for sustainable development, importing foreign technology is needed. The program has been extremely effective, teaching 22,000 farmers how to utilize drip irrigation technology. The program also prides itself on successfully engaging the government and private sector in its saving of nearly 4,000,000 m³ of water, increasing the yield of mango and rice by 25% and 5% respectively (SAF 2025). Additionally, another element of this program was creating “Farmers’ Hubs,” which are centers to improve coordination, training, and market access for the farmers implementing these technologies. The SAF highlighted that hubs lead to 283 new jobs to be created, the majority of which are female (SAF 2025).

The second intervention was significantly more market-based, promoting risk mitigation measures for climate change adaptation in agriculture. The project, which ran from 2018 to 2022, was funded by the Swiss Agency for Development and Cooperation and the Syngenta Foundation for Sustainable Agriculture. The program's main objective was to pilot and expand index-based crop insurance and other risk-mitigation tools for farmers growing rice, potato, and maize in Bangladesh. The program developed weather-index insurance products, allowing small farmers to get hyper-specific climate insurance. The intervention was extremely successful, ensuring over 400,000 farmers, 25% of whom were women. Nearly 150,000 received payouts totaling roughly 700,000 United States Dollars (SAF 2022).

While both significant and impressive development interventions in the realm of sustainable agriculture, the introduction of water-efficient technologies was slightly more

effective, as, in actuality, the programs amplify the effects of the technologies with social programs. The education and implementation of drip irrigation represent the technological element. The creation of the “Farmers’ Hubs” creates jobs and allows for further communication, education, and coordination for future market success for the farmers. Conversely, the second intervention, the risk-promoting risk mitigation measures, only utilizes the market-based approach of tailored climate insurance for rice, potato, and maize farmers.

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Climate Resilient Infrastructure: IGO-Based Loan Development vs NGO-Based Development

In this post, I compare two initiatives in Bangladesh attempting to promote sustainable infrastructure. The first project, “Bangladesh: Coastal Towns Climate Resilience Project,” is funded by the Asian Development Bank (ADB), an Intergovernmental Organization (IGO), and the government of Bangladesh. The second project, “Climate Resilient Housing: An Adaptation Initiative Toward a Sustainable Future in Coastal Bangladesh,” is funded primarily by the Bangladesh Rehabilitation Assistance Committee (BRAC), a Non-Governmental Organization (NGO) focused on the development of Bangladesh.

Sustainable infrastructure is an example of an incredibly important adaptation-based policy for addressing climate change. In Bangladesh, all 8.2 million coastal households are currently exposed to some disruption of essential infrastructure services because of the effects of climate change (Adshead et al. 2024). These disruptions will affect upwards of 35 million people (Adshead et al. 2024). Both of the infrastructure projects I will discuss in this post are in some way focused on providing shelter that is resistant to extreme weather that has increased in intensity due to climate change. Li et al. found that rapid intensification events, periods when a cyclone’s maximum sustained wind speed increase is at least 30 knots in 24 hours, have increased in offshore areas within 400 km of coastlines, with the count tripling from 1980 to 2020 (Li et al. 2023). Specifically, Balaguru et al. found that the intensity of major tropical cyclones in the Bay of Bengal, for which Bangladesh has a large section of coastline, has dramatically increased from 1981 to 2010. The paper links the increase to climate change and larger environmental changes (Balaguru et al. 2014).

The ADB-funded project, “Bangladesh: Coastal Towns Climate Resilience Project,” aims to financially support the Bangladesh government, allowing it to strengthen and enhance disaster resilience in 22 vulnerable coastal towns by improving climate-resilient infrastructure and enhancing resilient livelihoods. As of June 2025, the most up-to-date project report I could find, 51% of proposed cyclone shelters had been built, 83% of suggested stormwater drainage systems had been constructed, and 41% of proposed roads had been altered to include the stormwater drainage (Asian Development Bank 2025). In the 22 vulnerable coastal towns, 16 shelters had been made, and over nearly 30 km of stormwater drainage had been constructed. The BRAC-funded project, “Climate Resilient Housing: An Adaptation Initiative Toward a Sustainable Future in Coastal Bangladesh,” has successfully developed micro-infrastructure, opening 35 climate-resilient houses in four coastal districts: Barguna, Satkhira, Bhola, and Patukhali. BRAC estimates that each house, which can accommodate 35-45 people in the event of a disaster, costs approximately 12,000 USD to build. Therefore, provided these estimates are correct, the BRAC project has cost approximately 420,000 USD and has built shelter for upwards of 1600 people in the event of climate-based natural disasters (Kibria 2024; UNEP Copenhagen Climate Centre 2024).

These two interventions both represent success stories for Bangladesh and the creation of sustainable infrastructure. However, the structure and scale of the interventions vary based on the actors funding the projects. The ADB-funded program is structured as a development loan to the Bangladesh government. The term of the loan, which was signed in 2022, provides the government 25 years to pay the full loan back with an initial 5-year grace period in which no payments need to be made (Asian Development Bank 2022). A development loan, compared to NGO funded projects, allows for a much larger scale. The ADB-funded project has a budget near

300,000,000 USD compared to the estimated 420,000 USD spent on the disaster structures in the BRAC project. BRAC is not expecting and is never going to get paid back for the money it spends building the climate-resilient infrastructure. The NGO is providing a more pure form of aid, whereas the ADB is simply providing the government of Bangladesh with the means to improve its own infrastructure in the short run without having to immediately deal with the cost of the construction. These two different approaches to development have their own strengths and weaknesses. The loan structure allows for larger-scale projects and more lofty goals, but comes with the bureaucratic burdens of working with IGOs and the national government of Bangladesh. Whereas the NGO driven environmental development must be more focused and cost-effective, but able to help specific communities and still have tangible impacts. In this post, I have detailed that both approaches to developing climate-resilient infrastructure work. However, with the understanding that 8.2 million coastal households are at risk of losing key services due to climate disasters, large-scale projects like the one funded by ADB are needed (Adshead et al. 2024). While the more local and effective efforts of NGOs like BRAC are important, in order for the large-scale change of the country, larger-scale government projects are required.

Citations:

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The Role of Education in Climate Action: Community Based vs School Based Climate Education in Bangladesh

In this post, I will discuss the importance of education in climate change action in Bangladesh. I will highlight the important role educators and indigenous communities must play in this education. I will then examine an intervention in which there was a randomized cluster intervention in schools demonstrating the effectiveness of climate education in Bangladesh.

Teachers are fundamental to children's education. Chowdhury, Ahmed, and Ahmed's paper examined how teachers in Sylhet, Bangladesh, perceived climate change differed depending on whether they taught at a public or private school. Perception of climate change was examined through questions of importance, understanding of causes and impacts, and belief in mitigation. The paper surveyed 95 educators. It found that most teachers, both public and private, recognized that climate change was an important issue, a threat to Bangladesh, and human-caused. Interestingly, the study found that public school teachers tended to show a slightly stronger awareness of climate change (Chowdhury et al. 2021). While learning, teachers characterize issues for students.

The Kabir et al 2015 study, published in PLOS ONE, is an example of a school-based climate change education intervention in Bangladesh. The intervention was a cluster randomization intervention trial attempting to measure the effectiveness of using a school manual on climate change and health protection, which was endorsed by Bangladesh's national curriculum and textbook board, to teach secondary school students about climate change. Their intervention took place in 60 schools across 3000 plus year seven students. Schools were randomly assigned either to the intervention or control group. All students took a pre-test to measure baseline understanding and then a post-test six months later. The group that received the

additional education had a 17.42% higher post-test score as compared to the control group. Kabir et al. conclude by stating that the educational initiative was very effective at improving kids' knowledge and understanding of climate change (Kabir et al. 2015).

In Bangladesh, with the rise of natural disasters, climate change will continue to be an issue that needs to be connected to the classroom. Luetz and Sultana explain how climate change education is a key to successful disaster preparation and resilience in Bangladesh. Their chapter explains that currently in Bangladesh, access to education is uneven and directly affects people's ability to be prepared for, respond to, and recover from the growing issue of climate-caused natural disasters (Luetz and Sultana 2019). They highlight that poverty is a massive barrier to education and, therefore, a lack of preparation for disasters. Families in vulnerable, poor areas prioritize immediate survival and income, which tends to pull children out of school and into work. The article's key point is that in addressing the need for greater climate education, governments can not simply add or increase the discussion of climate change in curricula. Instead, development actors must also work towards removing barriers to access education (Luetz and Sultana 2019).

One idea on how to remove barriers to education is to conduct educational initiatives within the indigenous communities. Datta's 2024 paper argues that relationality should be the basis of climate education. The paper defines relationality as the relationships shared between land, people, culture, and non-human life in an environment (Datta 2024). The article discusses how both the Chittagong Hill Tracts and specifically the Laitu Khyeng Indigenous community are at extreme risk of biodiversity loss and land degradation due to extreme weather events and climate change. The paper discusses how practices like land-based walks with elders, land-based story listening, and recalling can be effective tools to frame environmental education. Indigenous

knowledge is interwoven with culture, environment, and the community. Therefore, these practices can serve as an effective tool for climate education outside the traditional school setting, which I mentioned earlier, poses barriers to entry.

In this post, I have attempted to provide a grand overview of the importance of climate education and a few examples of how it is currently taking place in Bangladesh. Climate change education is fundamental to climate action as a whole, as a better-informed populace will lead more people to care for the environment and want policy change that will improve the climate. Both interventions I discussed, the classroom setting and the indigenous community setting, are necessary; they both could be improved. Had the school-based education been pitched for younger children, the impacts may have been large and may have reached more children. The most effective intervention would be one that could mix the indigenous and community-based nature of relationality with some basic scientific understanding. This theoretical intervention would be community-based, so effective for children in and out of schools, while also tying historical culture to modern scientific understanding of climate change.

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Solar Adaptation in Bangladesh: A Comparison of Individual vs Community Government Funded Solar Initiatives

In this post, I will examine two different government-run projects to promote the use of renewable energy in Bangladesh. Both of these projects highlight renewable energy, specifically solar energy, not exclusively for their green impacts but instead for their practicality for distributing energy and power for those who lack access currently. In these discussions, I have yet to talk about the role of SDG7: Affordable and Clean Energy in environmental governance. SDG 7, specifically target 7.2, “by 2030, increase the share of renewable energy substantially in the global energy mix,” plays a massive role in creating greener and more equitable development and therefore is key in any conversation around sustainable development (United Nations). Both of the programs I am going to discuss in this post, Bangladesh solar homes systems (SHS) program and Solar mini-grids, are government led founded through a combination of grants and development loans from intergovernmental organizations (IGOs), foreign aid programs like USAID, and the government of Bangladesh itself (Cabraal et al. 2021; World Bank 2024).

The Bangladesh SHS program is a nationwide off-grid electrification program scheme aiming to enable rural households to purchase solar photovoltaic (PV) systems for their home as a replacement for kerosene lighting (Hellqvist and Heubaum 2023). The program ran from 2003 to 2018, and it represents a healthier and more environmentally friendly option for households in which direct power grid connection would be extremely difficult and expensive due to location, geography, and existing infrastructure (Hellqvist and Heubaum 2023). Most of the over 680,000,000 USD budget was in the form of development loans from the World Bank and Asian Development Bank, valued at approximately 415,000,000 and 90,000,000 USD, respectively. Families who were purchasing these solar PV systems for their homes were buying them at a

reduced price, initiating with a down payment of approximately 15% and then paying the rest in installments over the next one to three years (Cabraal et al. 2021). The SHS program sought to match the cost of the solar PV systems initiation to the cost of the kerosene alternative. The program structured the installation cost for the solar PV system to be approximately 4.50 USD a month to match the average amount spent on kerosene for lighting (Clean Energy Solutions Center 2013). However, occasionally, the solar PV system could cost more than the alternative (End User Subsidies Lab).

The solar mini-grids project is an Infrastructure Development Company Limited (IDCOL) project aimed at building mini power grids powered by solar, or often solar-diesel hybrid, to provide low-emission access for households and businesses in communities that are off-grid or very remote. As of 2021, the program had installed 26 of these solar-diesel hybrids (Aziz and Chowdhury 2021). The IDCOL is a state-owned infrastructure company; therefore, these mini-grids are an example of direct state-funded environmental development projects in solar technology. The Aziz and Chowdhury 2021 study found that, for the 21 mini-grids they examined, they were generally cost-effective, but often they were inefficient because demand was overestimated (Aziz and Chowdhury 2021).

Both of these initiatives represent positive investment in solar energy by the Bangladesh government, attempting to connect people with poor or no access to the grid with electricity. Where these two programs dramatically differ is in scale. The SHS program is attempting to create individual solutions for each household, whereas the mini-grids project is attempting to help entire communities that lack reliable electricity. Moreover, the systems the SHS program is pushing are only strong enough to power lighting systems and very small appliances, whereas the mini-grids can power larger appliances and industrial loads (Rahman 2020). They also differ

in terms of who is paying the development loans back. In the case of the SHS program, individual households are buying the solar PV systems at a reduced price and paying with a one to three-year loan, whereas the IDCOL, and by extension the government of Bangladesh, is funding and building the mini-grid projects. Because the mini-grids project is not for individual households, it is not coming at the expense of the individual, and rather, the government is fronting the cost as it's seen as infrastructure improving the state. Furthermore, the environmental impact of these projects must be considered. The SHS project is transitioning households away from carcinogenic kerosene lamps towards fully sustainable solar energy (Lam et al. 2012). However, these are local individual solar systems, so there will be a large impact of making and replacing the batteries every few years. On the other hand, the mini-grids are for full communities and therefore, fewer systems are needed to reach more people. But these systems are not purely solar and are only aimed at low-emission energy; the majority examined in the Aziz and Chowhury study were solar-diesel hybrid systems (Aziz and Chowhury 2021).

While both of these solar development projects are positive, the mini-grid program seems as if it would be more effective due to its larger scale. Since the program simultaneously provides access to the grid to more people while also being able to power larger devices and handle industrial loads, it represents the more comprehensive of the two programs. In an ideal world, and possibly in the future, the grids should be fully solar and aimed at zero-emissions energy. Overall, both of these projects represent a commitment by the Bangladesh government towards environmental governance, solar energy, and SDG target 7.2.

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Biogas vs Biomass: Non-solar sources of renewable energy for Bangladesh

In this post, I am going to discuss two sources of renewable energy that Bangladesh currently utilizes to some extent: biogas and biomass. I will also be examining peer-reviewed studies by Hasan et al. and Bhattacharyya that evaluate the effectiveness of these renewables in Bangladesh. This will be building off of my discussion around SDG 7: Affordable and Clean Energy in environmental governance, and specifically target 7.2, “by 2030, increase the share of renewable energy substantially in the global energy mix,” (n United Nations). Both of these renewables currently exist to some extent in Bangladesh programs implemented by the Infrastructure Development Company Limited (IDCOL) and partially funded by the World Bank (IDCOL, “Other Renewable Energy Projects”; World Bank 2024).

First, biogas is a renewable fuel gas composed mainly of methane and carbon dioxide. That description does not immediately call to mind renewable energy, but it originates from organic material, like animal manure and food waste, that is broken down without oxygen. In Bangladesh, this typically entails household or small farms to produce gas for cooking and occasionally for electricity generation. It is important to note that electricity generation is more common at larger commercial sites (Hasan et al. 2022). In 2006, the IDCOL implemented domestic biogas programs in Bangladesh. They financed over 50,000 small-scale biogas plants throughout the country with 38 partner organizations (IDCOL, “Biogas and Bio-fertilizer Program”). The IDCOL has also financed nine biogas-based power plants with aid from the World Bank and USAID. These power plants are much larger than the individual small-scale plants mentioned above. Their largest of these power plants has a max capacity of 400kW. Hasan et al.’s 2020 piece examined the strengths and barriers to the implementation of biogas technology in Bangladesh. The piece served stakeholders across the biogas ecosystem. The

largest barriers it identified were high initial investment costs as well as low awareness and understanding of the technology and its benefits (Hasan et al. 2022). The key strength the article noted was efficient utilization of waste (Hasan et al. 2022). The core idea behind biogas is that it converts existing negative externalities tied to agriculture into renewable gas and energy. When farming manure is going to be created, and have a negative impact on the environment. Treating the manure and turning it into a gas to power a stove or energy source turns the once negative consequence of farming into a cheap, after initial setup, energy.

Second, biogas renewable energy refers to the utilization of plant organic materials as an energy source. Bangladesh is making use of rice husks, a byproduct of farming, and converting them into usable energy through thermochemical processes (Bhattacharyya 2014). Currently, the IDCOL has financed two rice husk gasification plants, one in Gazipur with a max capacity of 250 kW and the other in Thankurgaon with a max capacity of 400kW. These plants were funded with significant aid from a World Bank grant, making up approximately 60% of the Gazipur plants' funding (Khan 2008). Bhattacharyya 2014 examines the viability, particularly from a financial perspective, of off-grid electricity being supplied by rice husk biomass. The piece concludes that low demand tends to produce low utilization. There only tends to be large amounts of rice husk in the time directly after harvest, and therefore, there exists a possibility that the technology is not continuously utilized. Moreover, Bhattacharyya highlights that the high cost of construction implementation of the technology has relied on large grants and subsidies by outside parties, as seen in the case of the World Bank and the Gazipur plant (Bhattacharyya 20214).

While these systems are all very positive and examples of sustainable development by turning waste into gas and/or electricity, the more local and individual small-scale biogas

facilities seem to be the most effective. The biomass production seems to be extremely reliant on outside funding and has unbalanced usage timed specifically around the harvest. Conversely, manure and other biomass runoffs exist year-round. The small-scale biogas systems, that transition the biomass into gas for stoves or small amounts of electricity, provide a fantastic example of small-scale sustainable development, aiding at the individual level. Looking overall at all of the sustainable energy interventions I have discussed, biomass, biogas, SHS, and solar-diesel mini grids all have pros and cons. I believe that a combined focus on the larger-scale energy project with the solar-diesel mini grids and a more individually focused biogas solution for stoves would provide Bangladesh with the most effective pathway towards target SDG 7 and specifically target 7.2.

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Financing Small-Scale Development and Climate Shock Response: Effectiveness of Microcredits on Sustainable Development in Bangladesh

Many of the previous interventions that I have discussed have been implemented through structured microcredit or microfinance programs. In this post, I will discuss how these programs are advancing sustainable development in Bangladesh. I will discuss the effectiveness of microcredit as a funding mechanism for Solar Home Systems (SHS) and the trends that make families more likely to use microcredit.

I first want to contextualize microcredits as a whole in Bangladesh. The Grameen Bank of Bangladesh is considered a pioneer in using microcredit as a form of development aid, specifically through group-based lending agreements. In these agreements, lenders are grouped together and held liable collectively, with a formal structure to encourage loan repayment (Morduch 1999). There have, however, been large downsides to microcredit, as they have been shown to lead to gender based violence and community humiliation (Ali 2014). Specifically, the Ali article shared anecdotes in which women directly face physical and emotional violence in response to failing to repay their loans. In one example, women were suggested to sell their bodies to pay back the loans (Ali 2014).

As I mentioned in Post 5, Solar Adaptation in Bangladesh: A Comparison of Individual vs Community Government-Funded Solar Initiatives, the SHS program required individuals to pay for the solar panels and their installation. Often this was accomplished through microcredits. Khan et al.'s 2019 article discusses trends among people who opted to use microcredit to finance solar panels. The paper reports that 63% of rural households in the study area installed SHS with the aid of microcredit. Interestingly, female-headed households were more likely to use micro-finance for the SHS abode than male-headed households. The loans discussed in the Ali

2014 paper were all given to women. While this can be viewed as a great step for gender based development, it can also be criticized because often the motivation for loaning to women was that they were less likely to disappear and easier to control than men, meaning that they were given this opportunity because of their low place in society. My discussion of sustainable development so far has overlooked that sustainable alternatives are often more costly than their unsustainable alternatives. Therefore, in individual-level interventions such as SHS, finance is a key part of the intervention. Whether you have access to microcredit directly affects your ability to participate in SHS and support sustainable development. Khan et al.'s paper highlighted trends in which characteristics enabled people to obtain easier access to microfinance to install the SHS. To make the SHS program more effective, it would be beneficial to improve the microcredit system that supports people in installing solar panels.

Jordan's 2021 article discusses whether microcredit can and is used as a response to climate shocks in Bangladesh. Specifically, the piece asks whether microcredit supports adaptation rather than merely coping with the disaster. The study surveyed 172 people and found that they often use microcredit as part of a broader portfolio of borrowing arrangements after climate shocks. Importantly, Jordan 2021 found that the survey results suggest that microcredit primarily supports coping and incremental adaptation, with a very limited ability to address deeper drivers of climate vulnerability (Jordan 2021). These findings are logical because larger, preexisting vulnerabilities to climate change are difficult to address through the small-scale individual actions that microcredit appears to support effectively.

Overall, microcredit is an effective tool for motivating individual small-scale sustainability. However, it would be interesting to examine the effectiveness of microcredit specifically designed to fund sustainable development projects. I was unable to find a program of

this nature that currently exists in Bangladesh; however, the loan structure around the SHS is most similar to this idea. To reiterate what I stated in my post on solar adaptation, SHS were traditionally financed with a down payment of approximately 15%, followed by payments, often through microcredit, in installments over the next one to three years (Cabraal et al., 2021). This example demonstrates how providing friendly loans to people who use the money to improve their livelihoods while simultaneously aiding the nation's sustainable development can have a significant positive impact.

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The Role of Gender In Bangladesh's Environmental Governance and Climate Action

A key element of any development goal or intervention I have yet to discuss is gender's role in the intervention. In this post, I will examine two examples of environmental governance and climate action that focus not only on SDG 13, Climate Action, but also on SDG 5: Gender Equality. I will examine two interventions aimed at increasing women's access and power in disaster risk reduction and relief. One program, a series of government policies focusing on making cyclone shelters and evacuation systems responsive to women's needs, focuses on providing equal access. In contrast, the other, Inclusive Community Disaster Risk Reduction and Management (IC-DRM), is a tangible example of action that focuses on placing women in decision-making roles in risk management.

The equal access initiative is a government-led program that focuses on improving access to cyclone shelters, evacuation systems, and disaster policies to meet women's needs better. Chowdhury et al. 2022 study, based on interviews with women who have needed to evacuate from coastal areas of Bangladesh in the past, found that women's shelter experiences were shaped by concerns around privacy, safety, health, and dignity (Chowdhury et al. 2022). The paper argues that these women's voices, stories, and opinions should be used to improve shelter design and management and advance gender equity in Bangladesh. The National Adaptation Plan of Bangladesh, a national development plan for 2023 to 2050, aims to implement these changes by repairing and rehabilitating cyclone shelters to ensure gender needs are met (Ministry of Environment, Forest and Climate Change, 2022). The Bangladesh Climate Change and Gender Action Plan, released in 2024, goes further, stating that women's leadership in the operation and maintenance of the shelters is necessary (Ministry of Environment, Forest and Climate Change 2024). While both of the aforementioned national plans are fantastic for gender

equality in disaster relief, I have been unable to find published, peer-reviewed information on their progress. These are large government documents that show that a discussion about the role of gender in climate action is underway in Bangladesh. Still, they do not yet demonstrate any tangible action.

The IC-DRM, however, is an example of tangible action being taken in Bangladesh. The intervention focused on bringing women and girls into local disaster planning and decision-making. The IC-DRM is a project in Bhola, a southern coastal city at major risk of cyclones and other climate change-caused natural disasters, that established two community-level action groups. The two groups, Ward Disaster Management Committees and Children and Youth groups, were created to encourage and increase the active participation, leadership, and decision-making power of women and girls in local disaster risk planning and preparedness (Plan International 2021). The UN Women report states that 1,600 plus women and girls across 27 groups in disaster-vulnerable areas are more actively participating in local disaster risk planning because of projects such as the Bhola program, and around 14,000 disaster-vulnerable women and girls are now actively participating in local disaster management committees (Plan International 2021; UN Women 2023).

These programs and government plans demonstrate the Bangladesh government's commitment to gender equality in climate action. They demonstrate both access-based governance, in the form of improving the cyclone structure to meet women's needs, and participation-based governance, in the form of including women in discussions within the IC-DRM example. However, they also highlight a broader problem with development as a whole: the need to produce documents that make grand promises but offer few policy examples to back them up. While the IC-DRM is a great example of gender equality in climate action, it is

not an example of infrastructure change, of which the National Adaptation Plan of Bangladesh and Bangladesh Climate Change and Gender Action Plan claim will take place in the context of cyclone shelters. Therefore, by my current estimate, Bangladesh is missing tangible examples of access-based governance in climate action, rather than government documents that simply claim it will happen in the near future.

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